



Weak Muller acceptance conditions for tree automata[☆]

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Abstract

Over the last decades the theory of automata on infinite objects has been an important source of tools for the specification and the verification of computer programs. Trees are more suitable than words to model nondeterminism and concurrency. In the literature, there are several examples of acceptance conditions that have been proposed for automata on infinite words and then have been fruitfully extended to infinite trees. The type of acceptance condition can influence both the succinctness of the language acceptors and the computational complexity of the decision problems. Here we consider, relatively to automata on infinite trees, two acceptance conditions that are obtained by a relaxation of the Muller acceptance condition: the *Landweber* and the *Muller-Superset* conditions. We prove that Muller-Superset tree automata accept the same class of languages as Büchi tree automata. Also, we show that for such languages the minimal Muller-Superset acceptor is at least as succinct as the minimal Büchi acceptor and, in some cases, it can be exponentially more succinct. Landweber tree automata, instead, define a class of languages that is not comparable with that defined by Büchi tree automata. The main result we prove is that the emptiness problem for this class of automata is decidable in polynomial time, and thus we extend the class of automata with a tractable emptiness problem.

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1. Introduction

Since its early days the theory of automata had an astonishing impact in computer science. Several models of automata have been extensively studied and applied to many fields. In the sixties, with their pioneering work, Büchi [2,3], McNaughton [17], and Rabin [18] enriched this theory by introducing finite automata on infinite objects. The connections between such automata and the formal theories have been fruitfully investigated and have originated automata-theoretic approaches to reduce decision problems in the field of mathematical logics to automata decision problems [9,13,21,22].

Automata on infinite words and trees turn out to be very useful for those areas of computer science where nonterminating computations are studied. They give a unifying paradigm to specify, verify, and synthesize nonterminating systems [13,22,23]. A system specification can be translated into an automaton, and thus, questions about systems and their specifications are reduced to decision problems in the automata theory. For example, the satisfiability of a specification can be reduced to checking for the nonemptiness of a language accepted by an automaton. Also, the correctness of a system with respect to a given specification can be rephrased as an instance of the language containment problem. It is thus important to study classes of automata for which nonemptiness is tractable and closure with respect to intersection holds.

In system modeling, trees are more suitable than words to model nondeterminism, which is also useful to model concurrent programs (nondeterministic interleaving of atomic processes). It is worth noticing that some concurrent programs, such as operating systems, communication protocols, and many control systems, are intrinsically nondeterministic and nonterminating. Moreover, by using trees we can express the existential path quantifier, and thus we are able to express lower bounds on nondeterminism and concurrency. This feature turns out to be greatly helpful in applications such as program synthesis [4,5].

In the literature, several acceptance conditions on infinite words have been fruitfully extended to infinite trees, such as Büchi, Muller, and Rabin conditions [21]. The kind of acceptance condition can influence the succinctness of the model, the computational complexity of the decision problems, and the closure properties of the accepted languages. While for Büchi tree automata the emptiness problem is decidable in polynomial time, for Rabin tree automata it is NP-complete. On the other hand, Büchi tree automata are not closed under language complementation, while Rabin tree automata are. Since Rabin tree automata are strictly more expressive than Büchi tree automata (in terms of the class of accepted languages), it is worth searching for new models of automata with interesting closure properties and tractable decision problems.

For automata on infinite objects, the acceptance is defined with respect to the set of states which are visited infinitely often while reading the input. For example, for a Büchi tree automaton, some of the states are accepting and acceptance is granted when on all paths of a tree at least an accepting state is visited infinitely often. For Muller tree automata, the accepting states are given as a collection of sets of states with the meaning that on each path of a tree the set of states that repeat infinitely often is exactly one of the accepting sets. In this paper, we study two new acceptance conditions for tree automata: *Landweber* and *Muller-Superset* acceptance conditions. They are obtained by relaxing the Muller condition

in the following way. The Landweber condition requires that, on each path of the input tree, the set of states that are visited infinitely often is contained in one of the accepting sets, that is, the states that repeat infinitely often need to belong to an accepting set but do not need to be exactly all the states of an accepting set. The Muller-Superset condition, instead, requires the opposite, i.e. on each path of the input tree, one of the accepting sets is contained in the set of states that are visited infinitely often, i.e. the states that repeat infinitely often are at least all the states of an accepting set.

With Landweber tree automata, we extend to infinite trees the acceptance condition introduced by Landweber in 1969, relatively to deterministic finite automata on infinite words [14].¹ Hossley studied the nondeterministic version of such automata [10]. This acceptance condition has also been considered for timed automata, for the verification of real-time systems [1]. Here, we study both the deterministic and the nondeterministic versions of Landweber tree automata.

For Landweber tree automata, we prove that the class of languages that are accepted by the deterministic model is strictly contained within the class defined by the nondeterministic one. We compare these classes to those accepted by Büchi and Muller tree automata in both deterministic and nondeterministic paradigms, and in particular, we prove that the class of languages accepted by Büchi tree automata is not comparable with that accepted by Landweber tree automata. We also prove that both the introduced classes are closed under intersection but not under complementation. Closure under union holds only for the nondeterministic paradigm. Our main result is that the emptiness problem for Landweber tree automata is decidable in polynomial time. From the above-mentioned results on the language comparison, we thus obtain a new class of tree languages with a tractable emptiness problem.

For Muller-Superset tree automata, we show that the class of accepted languages coincides with the class of languages accepted by Büchi tree automata, in both the deterministic and the nondeterministic versions. An interesting feature of this paradigm is that automata from this class can be more succinct than Büchi tree automata. We prove that for every language L accepted by a minimal Muller-Superset tree automata S , the language L is accepted by a minimal Büchi tree automaton B such that $Size(S) \leq Size(B) \leq 2^{O(Size(S))}$.

The rest of the paper is organized as follows: in Section 2, we give the definitions and recall results on the theory of finite automata on infinite trees. In Section 3, we study Muller-Superset tree automata and compare them to Büchi tree automata. In Section 4, we extend the Landweber acceptance condition to tree automata, and study the main closure properties and the comparison between deterministic and nondeterministic paradigms. Relationships among Büchi, Landweber, and Muller classes of languages are studied in Section 5. In Section 6, we prove that the emptiness problem for Landweber tree automata is decidable in polynomial time. Finally, we give a few remarks in Section 7.

¹ The Landweber acceptance condition is also known in the literature as generalized co-Büchi acceptance since it is dual to generalized Büchi acceptance [11,12]. While on words it captures the complement of Büchi accepted languages, on trees this does not hold [15].

2. Automata on ω -trees

In this section, we introduce the notation that is used in the rest of this paper. We also recall the definitions and the main results concerning Büchi, Muller, and Rabin tree automata [9,21].

Let Σ be an alphabet, an ω -word over Σ is a mapping from the set of nonnegative integers $\mathbb{N}$ into Σ , i.e. an infinite sequence of symbols over Σ . Let k be a positive integer and $\text{DOM} = \{0, 1, \dots, k-1\}^*$, we define an *infinite k -ary Σ -tree* t as a map $t : \text{DOM} \rightarrow \Sigma$. In the following, unless differently stated, an infinite k -ary Σ -tree will be referred simply as a tree. For each tree, the elements in DOM are the nodes of the tree and the empty word ε corresponds to the root. If u is a node of a tree, then ui is the i th child of u , for $i \in \{0, \dots, k-1\}$. Let $u, v \in \text{DOM}$, we say that u *precedes* v , denoted as $u < v$, if there exists an x such that $v = ux$.

Let $\pi \subseteq \text{DOM}$, π is *path* of a tree t if it is a maximal subset of DOM linearly ordered by $<$. Moreover, we define a *finite path* of t as a finite prefix of a path of t . For a path $\pi = u_0u_1 \dots$ of a tree t , the corresponding sequence of labels $t(u_0)t(u_1) \dots$ is an ω -word over Σ . If π is a path of a tree t , then t/π denotes the restriction of the function t to the set π . We say that a symbol $a \in \Sigma$ occurs in a tree t if there exists $u \in \text{DOM}$ such that $t(u) = a$. Moreover, we say that a occurs infinitely often in t/π if there exists an infinite number of nodes $u \in \pi$ such that $t(u) = a$. Let t be a tree and π a path of t , we denote the set $\{a \in \Sigma \mid a \text{ occurs infinitely often in } t/\pi\}$ by $\text{Inf}(t/\pi)$.

Given a tree t and a node $u \in \text{DOM}$, we define the *subtree* of t rooted at u as the tree t_u such that $t_u(v) = t(uv)$ for $uv \in \text{DOM}$. Let Σ be a finite alphabet, we denote by T_Σ^ω the set of Σ -valued trees. A language is a subset of T_Σ^ω . In the following we deal exclusively with binary trees ($\text{DOM} = \{0, 1\}^*$). All the results we obtain also hold for k -ary trees. According to the definition of a subtree, we use t_0 and t_1 to denote, respectively, the left and right subtree of t , while the 0th and 1th children of a node u are called, respectively, the left and the right children of u . Moreover, we denote by π_0 the leftmost path in a tree, that is $\pi_0 = \{0\}^*$, and by π_1 the rightmost one, i.e. $\pi_1 = \{1\}^*$.

Now we give some basic definitions from the theory of automata on infinite trees. We start by recalling some well-known acceptance conditions: Büchi, Muller, and Rabin automata. Other conditions have been considered in literature, such as the Streett condition [21] which defines a class of automata that turns out to be language equivalent to Muller tree automata.

Definition 1. A finite automaton on infinite trees (TA) is a tuple $A = \langle Q, \Sigma, \delta, Q_0, F \rangle$ where $Q \neq \emptyset$ is a finite set of states, Σ is an alphabet, $\delta \subseteq Q \times \Sigma \times Q \times Q$ is the transition relation, $Q_0 \subseteq Q$ is the set of initial states, and F is the acceptance condition.

A deterministic automaton on infinite trees (DTA) is a TA with $|Q_0| = 1$ and δ is a total function $\delta : Q \times \Sigma \rightarrow Q \times Q$. To describe the behavior of a TA, we recall the notion of run.

Definition 2. Let $A = \langle Q, \Sigma, \delta, Q_0, F \rangle$ be a TA and $t : \text{DOM} \rightarrow \Sigma$ be a tree. A run r of A on t is a tree $r : \text{DOM} \rightarrow Q$ such that $r(\varepsilon) \in Q_0$, and $(r(u), t(u), r(u0), r(u1)) \in \delta$, for all $u \in \text{DOM}$.

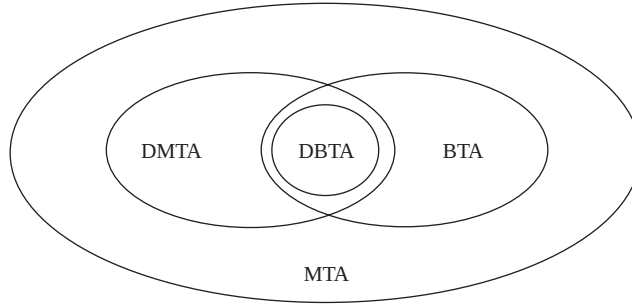


Fig. 1. Relationships among (D)BTA and (D)MTA.

We denote the set of runs of a TA A on a tree t by $Run_A(t)$. Clearly, if A is deterministic, then $|Run_A(t)| = 1$. Different classes of languages are obtained by defining different notions of *successful run*. A tree t is accepted by a TA A if there exists a successful run r of A on t . Moreover, with $T(A)$ we denote the language accepted by A , i.e. the set of accepted trees.

Büchi tree automata (BTA) were defined by Rabin [19] who originally named them *special automata*. A (D)BTA B is a (D)TA with a condition given by a set of accepting states $F \subseteq Q$, with the meaning that at least one accepting state occurs infinitely often on each path (*Büchi condition*). Thus, given a $t \in T_\Sigma^\omega$, a run $r \in Run_B(t)$ is successful if and only if for each path π of r , $Inf(r/\pi) \cap F \neq \emptyset$. We also denote the class of languages accepted by (deterministic) Büchi tree automata by (D)BTA.

The Muller condition is given by a collection of accepting sets of states F and requires that the states that occur infinitely often in a path form a set from F . Thus, a (deterministic) Muller tree automaton ((D)MTA) M is a (D)TA with acceptance condition $F = \{F_1, \dots, F_m\}$, with $F_i \subseteq Q$. Given a $t \in T_\Sigma^\omega$, a run $r \in Run_M(t)$ is successful if and only if for each path π of r , there exists a set $F_i \in F$ such that $Inf(r/\pi) = F_i$. We also denote the class of languages accepted by (deterministic) Muller tree automata by (D)MTA.

Finally, a (deterministic) Rabin tree automaton ((D)RTA) R is a (D)TA along with a set of pairs $F = \{(F_1, G_1), \dots, (F_m, G_m)\}$, where $F_i, G_i \subseteq Q$, for $i = 1, \dots, m$. Given a tree t , a run $r \in Run_R(t)$ is successful if and only if for each path π of r , there exists a pair $(F_i, G_i) \in F$ such that $Inf(r/\pi) \cap F_i \neq \emptyset$ and $Inf(r/\pi) \cap G_i = \emptyset$. We also denote the class of languages accepted by a (deterministic) Rabin tree automata by (D)RTA.

The following remark recalls the known relationships between the considered classes of tree automata [16,19,21]. Since the class of languages accepted by Rabin tree automata coincides with that accepted by Muller tree automata, when we compare the classes of languages we will refer only to MTA. Fig. 1 summarizes these known relationships.

Remark 1.

- (1) $DBTA \subset BTA \cap DMTA$;
- (2) BTA and DMTA are not comparable;
- (3) $BTA \cup DMTA \subset MTA$.

Languages	Ranking
$T_1 = \{t \in T_\Sigma^\omega \mid \exists \pi \text{ such that } a \notin \text{Inf}(t/\pi)\}$	BTA - DMTA
$T_2 = \{t \in T_\Sigma^\omega \mid \forall \pi, \text{ either } a \notin \text{Inf}(t/\pi) \text{ or } b \notin \text{Inf}(t/\pi)\}$	DMTA - BTA
$T_3 = \{t \in T_\Sigma^\omega \mid \forall \pi, a \in \text{Inf}(t/\pi) \text{ and } b \in \text{Inf}(t/\pi)\}$	DBTA
$T_4 = \{t \in T_\Sigma^\omega \mid a \notin \text{Inf}(t/\pi_0)\}$	(BTA $\cap$ DMTA) - DBTA
$T_5 = \{t \in T_\Sigma^\omega \mid t_0 \in T_1 \text{ and } t_1 \in T_2\}$	MTA - (BTA $\cup$ DMTA)

Fig. 2. Some tree languages and their classification.

In Fig. 2, we list some languages along with their ranking relatively to the classification illustrated in Fig. 1. For all these languages, we assume that $\Sigma = \{a, b\}$. We often refer to these languages in the following.

The ranking of T_1 and T_2 is proved in [21]. Directly from this result, it follows that T_5 is not in (BTA $\cup$ DMTA). From Remark 1, it follows that T_1 and T_2 are in MTA. Let M_1 and M_2 be two MTA accepting, respectively, T_1 and T_2 , and let M be an MTA that mimics M_1 on t_0 and M_2 on t_1 , for each tree t . Thus, $T_5 = T(M)$ and T_5 is in MTA - (BTA $\cup$ DMTA). To prove the ranking of T_3 consider the DBTA $A = \langle \{q_0, q_a, q_b\}, \Sigma, \{q_0\}, \delta, \{q_b\} \rangle$, where $\delta = \{(q_0, a, q_a, q_a), (q_0, b, q_0, q_0), (q_a, b, q_b, q_b), (q_a, a, q_a, q_a), (q_b, a, q_a, q_a), (q_b, b, q_0, q_0)\}$. The automaton A moves from a state $q \in \{q_0, q_b\}$ to the final state q_b anytime a sequence b^*aa^*b is read in input on a path. Thus, T_3 is in DBTA. To prove the ranking of T_4 consider the language of ω -words $L = \{t/\pi_0 \mid t \in T_4\}$. In [21] it is proved that L is accepted using the Muller paradigm, in both the deterministic and nondeterministic version of the automata, and also using the Büchi paradigm but only in the nondeterministic version. Thus, it is possible to construct a BTA and an MTA accepting T_4 by mimicking, respectively, the Büchi and Muller automata accepting L , on the path π_0 of each tree t given in input. On the other hand, there is no DBTA accepting T_4 . Indeed, suppose by contradiction that such an automaton B exists, then using B , it is possible to construct a deterministic Büchi automaton B' on ω -words recognizing L , just considering the left child in each transition of B . Thus, T_4 is in (BTA $\cap$ DMTA) - DBTA.

In the following remark, we summarize the closure properties of the above classes of automata and languages [16,19,21].

Remark 2.

- DMTA and DBTA are closed under intersection, but they are not closed under union and complementation.
- BTA is closed under intersection and union, but it is not closed under complementation.
- MTA is closed under intersection, union, and complementation.

In the following remark, we recall some known results on the decision problems of Büchi and Rabin tree automata.

Remark 3.

- The emptiness problem for BTA is decidable [19], and is LOGSPACE-complete for PTIME [22].
- The non-emptiness problem for RTA is NP-complete [6,18].

Given a set X , the size of X , denoted $Size(X)$, is defined either as the cardinality of X , if X is a set of states, or as $\sum_{X_i \in X} |X_i|$, if X is a collection of sets. The size of a TA $A = \langle Q, \Sigma, \delta, Q_0, F \rangle$ (denoted $Size(A)$) is defined as the sum of $Size(Q)$, $Size(\delta)$ and $Size(F)$. Let T be a language in BTA, we define the size of a description of T via the class BTA, denoted by $Size_{BTA}(T)$, as the minimum over the sizes of all BTA accepting T , that is, $Size_{BTA}(T) = Size(A)$ where A is a BTA such that $T(A) = T$ and for any other BTA A' such that $T(A') = T$, $Size(A) \leq Size(A')$. We refer to such a TA A as a minimal TA for T . In an analogous way we can define the size of a description of a language via any other class of languages.

3. Muller-Superset tree automata

In this section, we study the Muller-Superset acceptance condition, a new acceptance condition for tree automata obtained by using the superset operation, instead of the equality, in the Muller acceptance paradigm. We prove that Muller-Superset tree automata define the same class of languages as Büchi tree automata, in both the deterministic and non-deterministic cases. We also show that Muller-Superset tree automata are at least as succinct as Büchi tree automata and in some cases the gain can be exponential. We start by defining this class of automata.

A Muller-Superset tree automaton (STA) S is a TA with acceptance condition $F = \{F_1, \dots, F_m\}$, with $F_i \subseteq Q$. A tree $t \in T_\Sigma^\omega$ is accepted by an STA S if and only if there exists a run $r \in Run_S(t)$ satisfying the condition that for every path π of r , the inclusion $Inf(r/\pi) \supseteq F_i$ holds for some $F_i \in F$. Moreover, we also denote the class of languages accepted by (deterministic) Muller-Superset tree automata by (D)STA. In the next example we give a DSTA accepting the language $T_3 = \{t \in T_\Sigma^\omega \mid \forall \pi, a \in Inf(t/\pi) \text{ and } b \in Inf(t/\pi)\}$ listed in Fig. 2.

Example 1. Let $S = \langle \{q_0, q_a, q_b\}, \Sigma, \delta, \{q_0\}, \{\{q_a, q_b\}\} \rangle$ be a DSTA where $\delta = \{(q, x, q_x, q_x) \mid q \in Q \text{ and } x \in \Sigma\}$. A tree $t \in T(S)$ if and only if there exists a run $r \in Run_S(t)$ such that for all paths π of r , $\{q_a, q_b\}$ is a subset of $Inf(r/\pi)$, and thus if and only if for all paths π of t , $\{a, b\} \subseteq Inf(t/\pi)$ holds. Hence, we have that $T(S) = T_3$.

In the following theorem, we show that the classes DBTA and DSTA are equivalent. We prove this result by showing that for every DBTA that recognizes a language T , there exists a DSTA that recognizes the same language T and vice-versa. Besides, we show that for every language T accepted by a DSTA, $Size_{DSTA}(T) \leq Size_{DBTA}(T) \leq 2^{O(Size_{DSTA}(T))}$.

Theorem 1. *The classes DBTA and DSTA coincide. Moreover, for a given tree language T , $Size_{DSTA}(T) \leq Size_{DBTA}(T) \leq 2^{O(Size_{DSTA}(T))}$.*

Proof. To prove that $\text{DBTA} \subseteq \text{DSTA}$, we use a simple automata construction involving only the accepting set. Given a DBTA $B = \langle Q, \Sigma, \delta, Q_0, F \rangle$, we construct a DSTA $S = \langle Q, \Sigma, \delta, Q_0, F' \rangle$ where $F' = \{\{q_i\} \mid q_i \in F\}$. A tree $t \in T(B)$ if and only if there exists an $r \in \text{Run}_B(t)$ such that $\text{Inf}(r/\pi) \cap F \neq \emptyset$, for each path π of r , and thus if and only if there exists an accepting set $X \in F'$ such that $X \subseteq \text{Inf}(r/\pi)$. Hence, $t \in T(B)$ if and only if $t \in T(S)$. Notice that $\text{Size}(B) = \text{Size}(S)$, and if B is a minimal DBTA such that $T(B) = T$, then $\text{Size}_{\text{DBTA}}(T) = \text{Size}(B)$. Since $\text{Size}_{\text{DSTA}}(T) \leq \text{Size}(S)$, we get $\text{Size}_{\text{DSTA}}(T) \leq \text{Size}_{\text{DBTA}}(T)$.

Now, let us prove that $\text{DBTA} \supseteq \text{DSTA}$. Let $S = \langle Q, \Sigma, \delta, Q_0, F \rangle$ be a DSTA. Define H as the set $\{W \in 2^Q \mid \forall F_i \in F, F_i \not\subseteq W\}$ that contains only the sets of states W such that for each $F_i \in F$, the set W is either strictly contained in F_i or is not comparable with it. Let $B = \langle Q \times H, \Sigma, \delta', Q_0 \times \{\emptyset\}, Q \times \{\emptyset\} \rangle$ be a DBTA such that $((q, X), a, (q_1, X_1), (q_2, X_2)) \in \delta'$ if $(q, a, q_1, q_2) \in \delta$ and for $i = 1, 2$

$$X_i = \begin{cases} \emptyset & \text{if } \exists F_j \in F \text{ such that } F_j \subseteq X \cup \{q_i\}; \\ X \cup \{q_i\} & \text{otherwise.} \end{cases}$$

The automaton B enters an accepting state if and only if all the states of an accepting set $F_j \in F$ have been visited. Thus, for a tree t , we have that $t \in T(S)$ if and only if $t \in T(B)$. Since the number of sets in H is at most exponential in the size of Q , we have that $\text{Size}(B)$ is bounded above by $2^{O(\text{Size}_{\text{DSTA}}(T))}$. Thus, $\text{Size}_{\text{DBTA}}(T) \leq 2^{O(\text{Size}_{\text{DSTA}}(T))}$. $\square$

Given a DSTA S accepting a language T , using the above algorithm we can construct a DBTA B accepting T with an exponential blow-up in the size of the automaton S . In the next example, we show a matching lower-bound for this construction. In particular we define a class of languages and a corresponding Muller-Superset tree automaton S such that a minimal DBTA B accepting $T(S)$ is such that $\text{Size}(B)$ is at least exponential in $\text{Size}(S)$.

Example 2. Let $\Sigma = \Sigma_a \cup \Sigma_b$, where $\Sigma_a = \{a_1, \dots, a_n\}$ and $\Sigma_b = \{b_1, \dots, b_n\}$. Consider the language $T = \{t \in T_\Sigma^\omega \mid \forall \pi \text{ of } t, \{a_i, b_i\} \subseteq \text{Inf}(t/\pi) \text{ for some } i \text{ such that } 1 \leq i \leq n\}$. We show that T is accepted by a DSTA S with $\text{Size}(S) = 4n^2 + 4n$ and a minimal DBTA B accepting T is such that $\text{Size}(B) \geq 2^n$. Let $S = \langle Q, \Sigma, \delta, \{q_{a_1}\}, F \rangle$ be a DSTA with $Q = \{q_x \mid x \in \Sigma\}$, $F = \bigcup_{i=1}^n \{q_{a_i}, q_{b_i}\}$, and $\delta = \{(q, x, q_x, q_x) \mid q \in Q \text{ and } x \in \Sigma\}$. Trivially $T(S) = T$ and $\text{Size}(S) = 4n^2 + 4n$.

Now we prove by contradiction that $\text{Size}_{\text{DBTA}}(T) \geq 2^n$. Assume that B is a DBTA with less than 2^n states such that $T = T(B)$. Notice that if t is such that the labeling along each path is of type $(wb_i)^\omega$, where $w \in \Sigma_a^*$, $b_i \in \Sigma_b$ and w contains a_i , then t belongs to T . For a word $w \in \Sigma_a^*$, we define $\text{symbols}(w) = \{a_i \mid a_i \in \Sigma_a \text{ and } w = w'a_iw''\}$, that is the set of symbols from Σ_a which occur in w . Define the equivalence $\equiv$ over Σ_a^* as $w \equiv w'$ if and only if $\text{symbols}(w) = \text{symbols}(w')$. Clearly $\equiv$ has index 2^n . Given a word w , consider a tree t^w such that w is the prefix of the word labeling the leftmost path of t^w . Since B has less than 2^n states, there exist two words w and w' such that: $w \neq w'$ and, the runs of B on t^w and $t^{w'}$ reach, on the leftmost path, the same state after reading, respectively, w and w' . Thus, let $a_i \in \text{symbols}(w)$ and $a_i \notin \text{symbols}(w')$. Consider now a tree t whose leftmost path is labeled by $(wb_i)^\omega$ and a tree t' whose leftmost path is labeled by $(w'b_j)^\omega$ for some b_j such that $a_j \in \text{symbols}(w')$. Clearly both t and t' belong to T , thus, the corresponding runs r and

r' of B are accepting runs. Let t'' be a tree which differs from t' only in the leftmost path which is labeled by $(w'b_i)^\omega$. Such t'' does not belong to T but, by the above assumptions, we have that the only run of B on t'' is accepting, and thus we have a contradiction.

We end this section by showing that Büchi and Muller-Superset tree automata capture the same class of languages also in the nondeterministic case.

Theorem 2. *The classes BTA and STA coincide. Moreover, for a given tree language T , $\text{Size}_{\text{STA}}(T) \leq \text{Size}_{\text{BTA}}(T) \leq (\text{Size}_{\text{STA}}(T))^3$.*

Proof. To prove that $\text{BTA} \subseteq \text{STA}$, we can use the same construction as in the deterministic case. We recall that construction also proves $\text{Size}_{\text{STA}}(T) \leq \text{Size}_{\text{BTA}}(T)$.

Now, let us prove $\text{BTA} \supseteq \text{STA}$. Let $S = \langle Q, \Sigma, \delta, Q_0, F \rangle$ be an STA, where $F = \{F_1, \dots, F_n\}$. We number the states of F_h from 0 to $n_h - 1$, where $n_h = |F_h|$, according to any arbitrary order. Define a BTA B with $T(B) = T(S)$ in the following way. For each path, the automaton B nondeterministically guesses a set of accepting states F_h and checks that all the states from this set are visited infinitely often. Its states are triples $[q, h, i]$, where $q \in Q$, h is the index of the guessed set F_h , and i counts the visited states of F_h . More formally, let $B = \langle Q', \Sigma, \delta', Q'_0, F' \rangle$ the desired BTA with set of states $Q' = \{[q, h, i] \mid q \in Q, 1 \leq h \leq n \text{ and } 0 \leq i \leq n_h\}$, set of initial states $Q'_0 = \{[q, h, 0] \mid q \in Q_0 \text{ and } 1 \leq h \leq n\}$, accepting set $F' = \{[q, h, n_h] \mid q \in Q \text{ and } 1 \leq h \leq n\}$ and transition relation δ' defined as follows: for any transition $(q, a, q_1, q_2) \in \delta$, δ' contains tuples $([q, h, i], a, [q_1, h_1, i_1], [q_2, h_2, i_2])$ such that at least one of h_1 and h_2 is equal to h , and for $j = 1, 2$ we have that if $h_j \neq h$ then $i_j = 0$, otherwise if $h_j = h$ then

- $i_j = i + 1$, if $i < n_h$ and q is the i th state of the set F_h ,
- $i_j = 0$, if $i = n_h$, and
- $i_j = i$, in all the other cases.

By the above construction we have that the size of Q' is $\text{Size}(F) |Q|$, $|\delta'| = \mathcal{O}(n |\delta| \text{Size}(F))$ and $\text{Size}(F') = n |Q|$, where n is the cardinality of F . Thus, we get that $\text{Size}(B) \leq (\text{Size}(S))^3$, and $\text{Size}_{\text{BTA}}(T) \leq (\text{Size}_{\text{STA}}(T))^3$ holds. $\square$

4. Landweber tree automata

In this section, we introduce the *Landweber* acceptance condition, a new acceptance condition on infinite trees obtained by using set inclusion, instead of equality, in the Muller acceptance condition. We study for the corresponding class of languages the main closure properties and compare the classes of languages accepted by the deterministic and the nondeterministic Landweber tree automata. The considered model extends to trees the acceptance condition introduced by Landweber for automata on ω -words [14].

A *Landweber* tree automaton (LTA) L is a TA with acceptance condition $F = \{F_1, \dots, F_m\}$, where $F_i \subseteq Q$ for all $i = 1, \dots, m$. A tree $t \in T_\Sigma^\omega$ is accepted by L if and only if there exists a run $r \in \text{Run}_L(t)$ satisfying the condition that for every path π of r , the containment $\text{Inf}(r/\pi) \subseteq F_i$ holds for at least an $F_i \in F$. Moreover, we denote the class of languages accepted by (deterministic) Landweber tree automata by (D)LTA.

In the following example, we give a DLTA accepting the language $T_2 = \{t \in T_\Sigma^\omega \mid \forall \pi, \text{ either } a \notin \text{Inf}(t/\pi) \text{ or } b \notin \text{Inf}(t/\pi)\}$.

Example 3. Let $L = \langle \{q_a, q_b\}, \{a, b\}, \delta, \{q_a\}, \{\{q_a\}, \{q_b\}\} \rangle$ be a DLTA where $\delta = \{(q, a, q_a, q_a), (q, b, q_b, q_b) \mid q \in Q\}$. The automaton L changes its state to q_a (respectively q_b) whenever an occurrence of a (respectively b) is read. Thus, $t \in T(L)$ if and only if on each path there are either finitely many occurrences of b or finitely many occurrences of a , i.e. if and only if $t \in T_2$. Hence, $T_2 \in \text{DLTA}$.

We show now that, also for this acceptance condition, nondeterminism strictly increases the ability of the model as a language acceptor.

Theorem 3. $\text{DLTA} \subset \text{LTA}$.

Proof. To prove this result, we show that there exists a language in LTA which is not in DLTA. Recall the language $T_1 = \{t \in T_\Sigma^\omega \mid \exists \pi \text{ such that } a \notin \text{Inf}(t/\pi)\}$. We show that $T_1 \in \text{LTA-DLTA}$.

Let L be the LTA $\langle \{q_0, q_1, q_2\}, \{a, b\}, \delta, \{q_0\}, \{\{q_1, q_2\}\} \rangle$ such that δ is given by $\{(q_0, x, q_0, q_1), (q_0, x, q_1, q_0), (q_1, x, q_1, q_1), (p, b, q_1, q_2), (p, b, q_2, q_1) \mid x \in \{a, b\} \text{ and } p \in \{q_0, q_2\}\}$. The automaton L nondeterministically select a path in the input tree and checks if there is a finite number of a 's (this is done by guessing the last occurrence of a on the selected path). Thus, a tree is accepted if and only if it is possible to select a path with a finite number of a 's. Hence, $T_1 = T(L)$.

To complete the proof, suppose that there is a DLTA L accepting T_1 . Let $t \in T_1$ such that t_0 (i.e. the left subtree of t) is entirely labeled by a , and t_1 (i.e. the right subtree of t) is entirely labeled by b and let r be a successful run from $\text{Run}_L(t)$. Consider now a tree t' obtained from t by exchanging the two subtrees t_0 and t_1 , i.e. $t'_0 = t_1$ and $t'_1 = t_0$. Obviously, $t' \in T_1$. Denoting by r' a successful run of L on t' , replace t'_1 for t_1 in t , and call the obtained tree t'' . Clearly, $t'' \notin T_1$. Since L is deterministic, and r and r' are both successful runs, we have that replacing r'_1 for r_1 in r , we obtain a successful run of L on t'' . Thus, $t'' \in T(L)$ and hence we get a contradiction. $\square$

By standard automata constructions it is possible to prove the following.

Theorem 4. *LTA is closed under union and intersection, DLTA is closed under intersection but not under union.*

Proof. Let $L' = \langle Q', \Sigma, \delta', Q'_0, F' \rangle$ and $L'' = \langle Q'', \Sigma, \delta'', Q''_0, F'' \rangle$ be two LTA such that $T(L') = T'$ and $T(L'') = T''$. The LTA accepting $T' \cap T''$ is $L = \langle Q' \times Q'', \Sigma, \delta, Q'_0 \times Q''_0, F \rangle$, where $((q', q''), x, (q'_1, q''_1), (q'_2, q''_2))$ is in δ if and only if (q', x, q'_1, q'_2) is in δ' and (q'', x, q''_1, q''_2) is in δ'' , and $F = \{X' \times X'' \mid X' \in F' \text{ and } X'' \in F''\}$. Since this construction preserves the determinism, the closure under intersection holds also for DLTA. The result for the union in the non deterministic case, is easily obtained, in a very standard way, by considering a new LTA whose components are the union of the components of L' and L'' , assuming that $Q' \cap Q'' = \emptyset$. Non-closure of DLTA under union can be proved analogously

to the non-closure result under union of top-down deterministic tree automata [8]. Consider the language containing just one tree t with the root labeled by a and such that t_0 (i.e. the left subtree of t) is entirely labeled by a , and t_1 (i.e. the right subtree of t) is entirely labeled by b . Moreover, consider the language containing just one tree t' with the root is labeled by a , the nodes of t'_0 labeled by b , and the nodes of t'_1 labeled by a . Both these languages are in DLTA but their union cannot be recognized by a DLTA because any DLTA recognizing t and t' would accept also a tree entirely labeled by a . $\square$

Now, we introduce some notation that will be used to prove the non-closure under complementation of (D)DLTA. Let Σ be a finite alphabet and let $c \notin \Sigma$ be a new symbol, we denote by $T_\Sigma^{(c)}$ the set of trees with both infinite and finite paths valued on $\Sigma \cup \{c\}$ with the restriction that c cannot label internal nodes and all the symbols in Σ can only label internal nodes. Let $t, t' \in T_\Sigma^{(c)}$, we denote by $t \cdot_c t'$ the tree obtained from t by replacing each occurrence of c with t' (c -concatenation). For $t \in T_\Sigma^{(c)}$, the ωc -concatenation of t denoted by $t^{\omega c}$ is the tree in T_Σ^ω obtained by iterating the c -concatenation on t .

The following theorem states the non-closure under complementation of both the deterministic and nondeterministic Landweber tree automata.

Theorem 5. (D)DLTA is not closed under complementation.

Proof. To prove this result we show that there is a language in DLTA whose complement with respect to T_Σ^ω is not in LTA. In Example 3, we give a DLTA accepting T_2 . Consider now $T = \{t \in T_\Sigma^\omega \mid \exists \pi \text{ such that } a \in \text{Inf}(t/\pi) \text{ and } b \in \text{Inf}(t/\pi)\}$. Clearly, T is the complement of T_2 with respect to T_Σ^ω . Suppose that there exists an LTA L such that $T(L) = T$. Consider now a tree t such that all the nodes, except those in the path π_{i_0} , are labeled by b , and the labeling along π_{i_0} is of type $w(b^{n+1}a)^\omega$, where $w \in \Sigma^*$ and n is the number of states of L . Since $\text{Inf}(t/\pi_{i_0}) = \{a, b\}$, then $t \in T$. Let r be a successful run of L on t . Thus, there exists an $F_i \in F$ such that $\text{Inf}(r/\pi_{i_0}) \subseteq F_i$, i.e. there exists a natural number h such that for each $h' \geq h$, $r(0^{h'}) \in F_i$. Hence, there exists an $m \geq h$ such that $t(0^i) = b$, for each $m \leq i \leq m + n$, and there exist j and k , $m \leq j < k \leq m + n$ such that $r(0^j) = r(0^k) \in F_i$. We can decompose r (respectively, t) as follows:

- r' (resp. t') is obtained from r (resp. t) by replacing the subtree r_{0^j} (resp. t_{0^j}) with a sole node labeled by c .
- r'' (resp. t'') is obtained from r_{0^j} (resp. t_{0^j}) by replacing r_{0^k} (resp. t_{0^k}) with a sole node labeled by c .
- r''' (resp. t''') is r_{0^k} (resp. t_{0^k}).

Thus, $r = r' \cdot_c r'' \cdot_c r'''$ and $t = t' \cdot_c t'' \cdot_c t'''$. Directly from the above properties of r and t , we have that the run $r' \cdot_c r''^{\omega c}$ is a successful run of L on $t' \cdot_c t''^{\omega c}$. But for the choice of j and k this tree contains only paths with a finite number of a . Hence, we contradict the hypothesis that $T(L) = T$. $\square$

Notice that the above proof is sufficient to prove that also T_3 is not in LTA.

From the proofs given in this section, the ranking of T_1 , T_2 , and T_3 , relatively to Landweber tree automata, can be derived. In Fig. 3, we summarize these results.

Languages	Ranking
$T_1 = \{t \in T_\Sigma^\omega \mid \exists \pi \text{ such that } a \notin \text{Inf}(t/\pi)\}$	LTA - DLTA
$T_2 = \{t \in T_\Sigma^\omega \mid \forall \pi, \text{ either } a \notin \text{Inf}(t/\pi) \text{ or } b \notin \text{Inf}(t/\pi)\}$	DLTA
$T_3 = \{t \in T_\Sigma^\omega \mid \forall \pi, a \in \text{Inf}(t/\pi) \text{ and } b \in \text{Inf}(t/\pi)\}$	$2^{T_\Sigma^\omega}$ - LTA

Fig. 3. The classification of T_1 , T_2 , and T_3 with respect DLTA and LTA.

5. Language comparisons

In this section we compare the classes of languages (D)LTA with the classes (D)BTA and (D)MTA.

The following theorem states the results of the comparisons involving the class DLTA.

Theorem 6.

- (1) *DLTA and (D)BTA are not comparable.*
- (2) *$DLTA \cup DBTA \subset DMTA$.*

Proof. To prove part 1, we recall that from the table in Fig. 3 we have that $T_2 \in \text{DLTA}$. Since we also know from the table in Fig. 2 that $T_2 \notin \text{BTA}$, and from both the tables in Figs. 2 and 3 that $T_3 \in \text{DBTA-DLTA}$, we obtain that DLTA and (D)BTA are not comparable.

To prove part 2, we first show, with an easy automata construction, that $DLTA \subseteq DMTA$. Let $L = \langle Q, \Sigma, \delta, Q_0, F \rangle$ be a DLTA and $M = \langle Q, \Sigma, \delta, Q_0, F' \rangle$ be a DMTA where $F' = \bigcup_{F_i \in F} 2^{F_i} - \{\emptyset\}$. Directly from their definitions, L and M accept the same language, and since M differs from L only for the family of the final states, M is deterministic if and only if L is deterministic as well. From the relationship summarized in Fig. 1, we get that also $DBTA \subseteq DMTA$, and thus, $LTA \cup BTA \subseteq DMTA$. To prove the strict containment, we use the language $T = \{t \in T_\Sigma^\omega \mid t_0 \in T_2 \text{ and } t_1 \in T_3\}$. Recall that $T_2 = \{t \in T_\Sigma^\omega \mid \forall \pi, \text{ either } a \notin \text{Inf}(t/\pi) \text{ or } b \notin \text{Inf}(t/\pi)\}$ and $T_3 = \{t \in T_\Sigma^\omega \mid \forall \pi, a \in \text{Inf}(t/\pi) \text{ and } b \in \text{Inf}(t/\pi)\}$. We know that T_2 is not in DBTA, from the table in Fig. 2, and T_3 is not in DLTA, from the table in Fig. 3. It follows that $T \notin DLTA \cup DBTA$. Moreover, since T_2 and T_3 are in DMTA (see, table in Fig. 2), it is possible to build a DMTA accepting T . Trivially, the desired DMTA simulates, on each tree t , the DMTA accepting T_2 on the subtree t_0 , and the DMTA accepting T_3 on the subtree t_1 . Thus, we have that part 2 of the theorem holds. $\square$

Comparisons between LTA and the other classes are established in the following lemma.

Theorem 7.

- (1) *LTA and (D)BTA are not comparable.*
- (2) *LTA and DMTA are not comparable.*
- (3) *For any different choice of $X, Y, Z \in \{LTA, BTA, DMTA\}$, we have that $X \not\subseteq Y \cup Z$.*
- (4) *$LTA \cup BTA \cup DMTA \subset MTA$.*

Proof. Part 1 is a direct consequence of $T_3 \in DBTA - LTA$, and $T_2 \in LTA - BTA$ (see tables in Figs. 2 and 3). Part 2 is a direct consequence of $T_1 \in LTA - DMTA$ and $T_3 \in DMTA - LTA$ (see tables in Figs. 2 and 3). To prove part 3, we only need to consider three cases: (a) $LTA \not\subseteq BTA \cup DMTA$, (b) $BTA \not\subseteq LTA \cup DMTA$, and (c) $DMTA \not\subseteq BTA \cup LTA$. Consider the case (a). Let $T = \{t \in T_\Sigma^\omega \mid t_0 \in T_1 \text{ and } t_1 \in T_2\}$. Since $T_1 \notin DMTA$ and $T_2 \notin BTA$ (see Fig. 2), we have that $T \notin DMTA \cup BTA$. Moreover, since both T_1 and T_2 belong to LTA (see Fig. 3), it follows that $T \in LTA$. The proofs for cases (b) and (c) are analogous. In particular, for (b) we can use $T = \{t \in T_\Sigma^\omega \mid t_0 \in T_1 \text{ and } t_1 \in T_3\}$ and for (c) we can use $T = \{t \in T_\Sigma^\omega \mid t_0 \in T_2 \text{ and } t_1 \in T_3\}$. For part 4, we first observe that, using the same construction as that used in the previous theorem to prove that $DLTA \subseteq DMTA$, we can also prove that $LTA \subseteq MTA$. To prove the strict containment we use the language $T = \{t \in T_\Sigma^\omega \mid t_0 \in T_1, t_{10} \in T_2 \text{ and } t_{11} \in T_3\}$ (recall that t_0 is the left subtree of t , while t_{10} and t_{11} are subtrees of t starting, respectively, at the nodes 10 and 11). Since $T_1 \notin DMTA$ and $T_2 \notin BTA$ (see Fig. 2), and $T_3 \notin LTA$ (see Fig. 3), it follows that $T \notin DMTA \cup BTA \cup LTA$. On the other hand, since T_1, T_2 and T_3 are in MTA , it is easy to build an MTA accepting T . Trivially, for each tree t , the desired MTA simulates the MTA accepting T_1 on the subtree t_0 , the MTA accepting T_2 on the subtree t_{10} and the MTA accepting T_3 on the subtree t_{11} . $\square$

In the following theorem we complete the study of the relationships among the introduced classes of languages.

Theorem 8.

- (1) $DLTA \cap DBTA \cap DMTA \neq \emptyset$.
- (2) $((LTA \cap DMTA) - BTA) \neq \emptyset$, $((BTA \cap LTA) - DMTA) \neq \emptyset$ and $((BTA \cap DMTA) - LTA) \neq \emptyset$.
- (3) $((DLTA \cap BTA) - DBTA) \neq \emptyset$.
- (4) $((BTA \cap DMTA) - (DBTA \cup LTA)) \neq \emptyset$.

Proof. To prove part 1, we show that there exists a language accepted by a $DLTA$, a $DBTA$, and a $DMTA$. As a candidate, we consider the language $T = \{t \in T_\Sigma^\omega \mid \forall x \in \pi_0, t(x) = a\}$. Let $B = \langle \{q_a, q_b, q\}, \Sigma, \delta, \{q_a\}, F \rangle$ be a $DBTA$ such that $\delta = \{(q_a, a, q_a, q), (q_a, b, q_b, q), (q_b, x, q_b, q), (q, x, q, q) \mid x \in \Sigma\}$ and $F = \{q_a, q\}$, and L be a $DLTA$ obtained from B by considering as accepting set $F = \{\{q_a\}, \{q\}\}$. In both cases, a tree t is accepted if only a 's occur on the path π_0 of t . Hence, $T \in DLTA \cap DBTA$. To complete the proof, we just notice that, from the Remark 1, $DBTA \subseteq DMTA$, thus, T is also in $DMTA$. Part 2 directly follows from the classification of the languages T_1, T_2 and T_3 given in the tables in Figs. 2 and 3. Part 3 can be proved by using the language $T_4 = \{t \in T_\Sigma^\omega \mid a \notin Inf(t/\pi_0)\}$ listed in Fig. 2. Indeed, we know that T_4 is in BTA but not in $DBTA$ (see table in Fig. 2). To complete the proof, we just need to prove that T_4 belongs to the class $DLTA$. Let $L = \langle \{q_a, q_b, q\}, \{a, b\}, \delta, \{q_a\}, \{\{q_b\}, \{q\}\} \rangle$ be a $DLTA$ where $\delta = \{(p, a, q_a, q), (p, b, q_b, q), (q, x, q, q) \mid p \in \{q_a, q_b\} \text{ and } x \in \Sigma\}$. A tree t belongs to $T(L)$ if and only if there is a finite number of a 's on t/π_0 , hence, $T(L) = T_4$. For part 4, consider $T = \{t \in T_\Sigma^\omega \mid t_0 \in T_3 \text{ and } t_1 \in T_4\}$. Since $T_3 \notin LTA$ (see Fig. 3), and

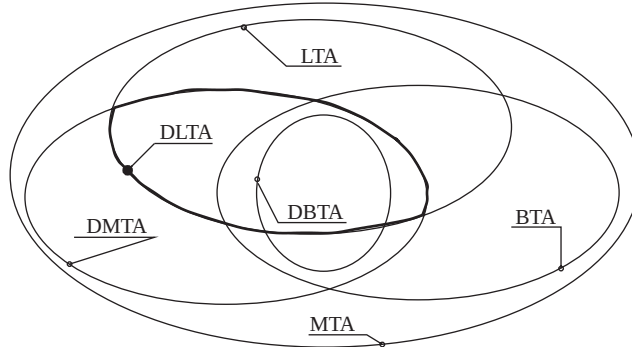


Fig. 4. Relationships between (D)BTA, (D)MTA and (D)LTA.

$T_4 \notin \text{DBTA}$ (see Fig. 2), we have that $T \notin \text{LTA} \cup \text{DBTA}$. Moreover, since T_3 and T_4 are in $\text{DMTA} \cap \text{BTA}$ (see Fig. 2), we have that $T \in \text{DMTA} \cap \text{BTA}$. $\square$

It is an open problem to determine whether $\text{DLTA} = \text{DMTA} \cap \text{LTA}$. Our conjecture is that this is the case. Our main motivation is that a language to be accepted nondeterministically by an LTA and deterministically by an MTA cannot contain trees that require nondeterministic guesses on the paths on which the automaton must check for a given property. Thus, nondeterminism can be only used along each path. Since every path is an infinite word and the nondeterministic and deterministic paradigms of finite automata on infinite sequences of characters with Landweber acceptance condition are equivalent [7], we expect our conjecture to hold.

Assuming that our conjecture is true the relationships among the introduced classes of languages is reported in Fig. 4. Notice that according to the results presented in this section, in the diagram each region within closed lines corresponds to a non empty subclass of languages.

6. The emptiness problem for LTA

In this section, we deal with the emptiness problem for Landweber tree automata. We prove that this problem is decidable in polynomial time. We recall that from Theorem 7 the class LTA is not comparable with BTA and is strictly contained in MTA. Thus, this result extends the class of languages with a tractable emptiness problem.

We observe that the Landweber condition can be directly translated to the Rabin condition by a very simple construction. Given a Landweber tree automaton $L = \langle Q, \Sigma, \delta, Q_0, F \rangle$ where $F = \{F_1, \dots, F_m\}$, a Rabin tree automaton accepting $T(L)$ is $R = \langle Q, \Sigma, \delta, Q_0, F' \rangle$, where $F' = \{(F_1, Q - F_1), \dots, (F_m, Q - F_m)\}$. By this construction and the result that any non-empty language accepted by a Rabin tree automaton contains a regular tree [20], we have the following lemma.

Lemma 1. *Each non-empty language accepted by a Landweber tree automaton contains a regular tree.*

Since the constructed automaton R is linear in the size of the starting automaton L , and the non-emptiness problem for Rabin tree automata is NP-complete [6,18], we get a nondeterministic polynomial time algorithm to solve the non-emptiness problem for Landweber tree automata. In the following, we indeed prove that checking for the emptiness of Landweber tree automata can be done in deterministic polynomial time and thus, under the theoretical assumption that “ $P \neq NP$ ”, it turns out to be simpler than for Rabin tree automata. Before proving the claimed result on the LTA emptiness problem we give some notation.

Consider an LTA $L = \langle Q, \Sigma, \delta, Q_0, F \rangle$, where $F = \{F_1, \dots, F_m\}$. A *finite portion* of a run r is a subtree r' of r such that all paths in r' are finite. We refer to the last nodes of all paths in r' as *leaves*. For a set of states $Q' \subseteq Q$, we define $Reach(Q')$ as the set of all the states from which there exists a finite portion of a run whose leaves are labeled with states in Q' . This set can be computed in time linear in $|Q'| + |\delta|$, by using a simple algorithm of backward reachability. In practice, such an algorithm starts by initializing a new set of states Q'' as Q' , then iteratively adds to Q'' all states $q \in Q$ such that there exists a transition $(q, x, q_1, q_2) \in \delta$ where $x \in \Sigma$, and both q_1 and q_2 are in Q'' . We end the iterations when no states from Q can be added to Q'' . Thus, we set $Reach(Q') = Q''$.

For a transition $e = (q, a, q_1, q_2) \in \delta$, we define the *graph induced* by e as the directed graph whose vertices, denoted $vertices(e)$, are q , q_1 , and q_2 , and whose edges, denoted $edges(e)$ are (q, q_1) and (q, q_2) . Given a set of transitions δ' such that for each $q \in Q$ there exists at most a transition of δ' from q , a graph induced by δ' is the graph having as set of vertices the set $\bigcup_{e \in \delta'} vertices(e)$ and as set of edges the set $\bigcup_{e \in \delta'} edges(e)$. If $\delta' \subseteq \delta$, this graph is also called a graph *embedded* in L . Given two sets $Q', Q'' \subseteq Q$, by $Accept(Q', Q'')$ we denote a set of states $q \in Q$ such that there exists a graph G embedded in L such that q is a vertex of G and along every maximal simple path of G starting from q either a state in Q'' or a cycle over only states from Q' are reached. Clearly, $Accept(\emptyset, P) = Reach(P)$ and $Accept(P, \emptyset)$ is $Reach(P')$ where $P' = \bigcup_{i=1}^k P_i$ is such that for any $i = 1, \dots, k$: $P_i \subseteq P$ and there exists a graph G_i that (i) is embedded in L , (ii) has P_i as set of vertices, and (iii) is a strongly connected component (i.e., from any vertex of G_i all the other vertices of G_i can be reached). The set $Accept(Q', Q'')$ can be computed by the algorithm in Fig. 5.

Lemma 2. *Algorithm ACCEPT computes $Accept(Q', Q'')$ in $\mathcal{O}(|Q'| |\delta| + |Q''|)$ time.*

The following theorem holds.

Theorem 9. *Given an LTA $L = \langle Q, \Sigma, \delta, Q_0, F \rangle$, the emptiness problem is decidable in $\mathcal{O}(|F| |\delta| |Q|^2)$ time.*

Proof. Consider a Landweber tree automaton $L = \langle Q, \Sigma, \delta, Q_0, F \rangle$, where $F = \{F_1, \dots, F_m\}$. A procedure to decide “ $T(L) \neq \emptyset$?” is given by the recurrence: $\Gamma_0 = \bigcup_{i=1}^m Accept(F_i, \emptyset)$ and $\Gamma_{j+1} = \bigcup_{i=1}^m Accept(F_i, \Gamma_j)$. Our decision algorithm outputs YES as soon as a set Γ_i such that $\Gamma_i \cap Q_0 \neq \emptyset$ is computed, and NO if the least fixed-point Γ of the above recurrence is computed and $\Gamma \cap Q_0 = \emptyset$. We observe that this fixed-point always exists since

```

Algorithm ACCEPT( $Q', Q''$ ).
 $P \leftarrow Q'$ 
repeat
   $P' \leftarrow P$ 
   $P \leftarrow \emptyset$ 
  for  $(q, a, q_1, q_2) \in \delta$  such that  $q \in P'$  do
    let  $p, p' \in \{q_1, q_2\}$  and  $p \neq p'$ 
    if  $(p \in Q''$  and  $p' \in P')$  or  $q_1, q_2 \in P'$  then
       $P \leftarrow P \cup \{q\}$ 
  enddo
until  $P' = P$ 
return  $\text{Reach}(Q'' \cup P)$ 

```

Fig. 5. Algorithm for computing $\text{itAccept}(Q', Q'')$.

the sets Γ_i grow monotonically within the set Q which is finite, and therefore, Γ is computed in at most $|Q|$ iterations. Thus, by Lemma 2, computing Γ requires $\mathcal{O}(m |\delta| |Q|^2)$ time. To complete the proof we need to show the correctness of our algorithm, i.e. we need to prove that $T(L)$ is not empty if and only if $\Gamma_i \cap Q_0$ is not empty for some i . The converse direction is a direct consequence of the definition of the set Γ and the definition of the Landweber acceptance condition. Consider now the forward direction. From Lemma 1 we have that $T(L)$ is not empty if and only if there exists a regular tree belonging to $T(L)$. Consider now a finite graph G corresponding to an accepting regular run of L (i.e., an accepting run on a regular tree). We claim that G must contain at least a strongly connected component over states from a given $F_i \in F$. To prove this we observe that if this is not the case, then there exists an infinite path π of G and thus of the corresponding regular tree, that does not satisfy the given acceptance condition. That is, for each $F_i \in F$, it holds that on π the set of states that repeat infinitely often is not contained in F_i . This property ensures that some of the vertices of such G correspond to states belonging to Γ_0 . By a similar argument, we can prove that at the i th iteration of the above algorithm at least a new state, among those corresponding to vertices of G , is added to Γ_i , and this is repeated until all such states are added. Since G corresponds to an accepting run of L , then also a state from Q_0 is eventually added to some Γ_i . Thus, we have shown that if $T(L)$ is not empty then for some i , also $\Gamma_i \cap Q_0$ is not empty, and this concludes the proof. $\square$

7. Conclusions

In this paper, we have studied two new acceptance conditions for tree automata: the Landweber and the Muller-Superset acceptance conditions. These conditions are relaxations of the Muller acceptance condition. We have shown that Muller-Superset tree automata capture the same class of languages as Büchi tree automata, and that automata from this class can be more succinct than Büchi tree automata. For Landweber tree automata we have studied both the nondeterministic and the deterministic paradigms, and compared the corresponding classes of languages to those characterized by Büchi and Muller tree automata. The class of languages accepted by Landweber tree automata is orthogonal to

that accepted by Büchi tree automata. For this class we have studied the main closure properties and the emptiness problem.

Our main result is that the emptiness problem for Landweber tree automata is decidable in polynomial time, and thus we enlarge the class of ω -tree languages that can be characterized by automata with a tractable emptiness problem. Besides the intrinsic interest in such a theoretical result, the emptiness problem of automata is closely related to decision problems in mathematical logics and also to the specification and verification of systems. Future developments of this research will take in account these aspects.

Finally, the Landweber acceptance condition naturally expresses the complement of the Büchi condition. Given a Büchi condition F its complementation is given by the Landweber condition $\{Q - F\}$, where Q is the set of states of the automaton. This suggests a very simple way of characterizing the complement of deterministic Büchi tree automata by nondeterministic Landweber tree automata with only one accepting set. The nondeterminism is needed to nondeterministically select a path to check for the violation of the Büchi condition. See [15] for more on the complementation of languages accepted by Büchi tree automata.

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